

Power Analysis

Clean Data GH2211 - 230327 & 230403

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Scripts

Load Libraries

```
library(tidyverse) #clean
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.3      ✓ readr      2.1.4
✓ forcats    1.0.0      ✓ stringr    1.5.0
✓ ggplot2    3.4.4      ✓ tibble     3.2.1
✓ lubridate  1.9.3      ✓ tidyr      1.3.0
✓ purrr      1.0.2
— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(flextable) #to make tables, not used in code
```

Attaching package: 'flextable'

The following object is masked from 'package:purrr':

compose

```
library(readxl)
```

Load the Raw data

Today I am working on two data sets from the 2211 nitrogen growth experiment from 2023-03-27 and 2023-04-03

```
df_230327 <- read_excel("~/PowerAnalysis-Research/power_analysis/CleanedData/GH2211_nitro
df_230403 <- read_excel("~/PowerAnalysis-Research/power_analysis/CleanedData/GH2211_nitro
```

- New names:
- `` -> `...8`
 - `` -> `...9`
 - `` -> `...10`
 - `` -> `...11`

Inspect Data

```
qflectable(head(df_230327))
```

block	pot #	height (mm)	SPAD (n=3)	date	GH_zone	comments
1	48	406	34.0	2023-03-27 00:00:00	3	
1	64	88	24.6	2023-03-27 00:00:00	3	
1	68	96	24.1	2023-03-27 00:00:00	3	
1	45	393	28.4	2023-03-27 00:00:00	3	
2	75	95	26.9	2023-03-27 00:00:00	3	
2	51	379	27.3	2023-03-27 00:00:00	3	

```
qflectable(head(df_230403))
```

block	pot #	height (mm)	SPAD (n=3)	date	GH_zone	comments	...8	...9	...10	...11
1	48	401	30.7	2023-04-03 00:00:00	3					
1	64	88	34.1	2023-04-03 00:00:00	3					
1	68	103	36.5	2023-04-03 00:00:00	3					
1	45	446	23.3	2023-04-03 00:00:00	3					
2	75	101	37.1	2023-04-03 00:00:00	3	high spad val, beginning new growth				
2	51	410	23.2	2023-04-03 00:00:00	3					

Clean Dataframes

Listing changes to be made to each data set in the order they will be implemented in the code chunk.

df_230327

Changes to be made

Data cleaning for df_230327

- 1. Rename SPAD (n=3) to spad
- 2. Rename pot # to pot_id
- 3. Convert height (mm) to centimeters
- 4. Rename height (mm) to height_cm
- 5. Add and convert date from MM-DD-YYYY to YYYY-MM-DD
- 6. Create a columns named row, inoc_status, abiotic, sub and leaf_count to match other data sets for consistencies.
- 7. Order columns, "date", "block", "row", "pot_id", "inoc_status", "abiotic", "height_cm", "leaf_count", "spad", "sub", "GH_zone".

```
df_230327_cleaned <- df_230327 %>%
  rename(spad = "SPAD (n=3)") %>%
  rename(pot_id = "pot #") %>%
  rename(height_cm = "height (mm)") %>%
  mutate(date = ymd("2023-03-27"),
    height_cm = height_cm / 10,
    row = NA,
    inoc_status = NA,
    abiotic = NA,
    leaf_count = NA,
    sub = NA) %>%
  select(date, block, row, pot_id, inoc_status, abiotic, height_cm, leaf_count, spad, sub, GH_zone)
qflectable(head(df_230327_cleaned))
```

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-03-27	1	48				40.6		34.0		3
2023-03-27	1	64				8.8		24.6		3
2023-03-27	1	68				9.6		24.1		3
2023-03-27	1	45				39.3		28.4		3
2023-03-27	2	75				9.5		26.9		3
2023-03-27	2	51				37.9		27.3		3

df_230403

Data cleaning for df_230403

- 1. Rename SPAD (n=3) to spad
- 2. Rename pot # to pot_id
- 3. Convert height (mm) to centimeters
- 4. Rename height (mm) to height_cm
- 5. Convert date from MM-DD-YYYY to YYYY-MM-DD
- 6. Create a columns named row, inoc_status, abiotic, sub and leaf_count to match other data sets for consistencies.
- 7. Order columns, "date", "block", "row", "pot_id", "inoc_status", "abiotic", "height_cm", "leaf_count", "spad", "sub", "GH_zone".

```
df_230403_cleaned <- df_230403 %>%
  rename(spad = "SPAD (n=3)") %>%
  rename(pot_id = "pot #") %>%
  rename(height_cm = "height (mm)") %>%
  mutate(date = ymd("2023-04-03"),
         height_cm = height_cm / 10,
         row = NA,
         inoc_status = NA,
         abiotic = NA,
         leaf_count = NA,
         sub = NA) %>%
  select(date, block, row, pot_id, inoc_status, abiotic, height_cm, leaf_count, spad, sub, GH_zone)
qflextable(head(df_230403_cleaned))
```

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-04-03	1	48				40.1		30.7		3
2023-04-03	1	64				8.8		34.1		3
2023-04-03	1	68				10.3		36.5		3
2023-04-03	1	45				44.6		23.3		3
2023-04-03	2	75				10.1		37.1		3
2023-04-03	2	51				41.0		23.2		3

Combine the data frames

```
df_combined_clean_327_403<- bind_rows(df_230327_cleaned, df_230403_cleaned)
qflectable(df_combined_clean_327_403)
```

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-03-27	1	48				40.6		34.0		3
2023-03-27	1	64				8.8		24.6		3
2023-03-27	1	68				9.6		24.1		3
2023-03-27	1	45				39.3		28.4		3
2023-03-27	2	75				9.5		26.9		3
2023-03-27	2	51				37.9		27.3		3
2023-03-27	2	61				45.9		25.9		3
2023-03-27	2	57				11.6		26.6		3
2023-03-27	3	62				10.3		27.8		3
2023-03-27	3	33				40.1		26.6		3
2023-03-27	3	52				9.4		24.5		3
2023-03-27	3	72				42.4		30.0		3
2023-03-27	4	70				7.7		28.8		3
2023-03-27	4	38				9.8		24.3		3
2023-03-27	4	74								3
2023-03-27	4	47				39.6		29.5		3
2023-03-27	5	73				41.8		28.8		3
2023-03-27	5	77				42.1		28.8		3

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-03-27	5		79			11.6		29.7		3
2023-03-27	5		39			7.6		22.4		3
2023-03-27	6		65			41.1		28.4		3
2023-03-27	6		22			7.6		25.1		3
2023-03-27	6		58			9.6		24.0		3
2023-03-27	6		66			43.5		30.6		3
2023-03-27	7		25			38.7		25.6		3
2023-03-27	7		41			10.2		30.6		3
2023-03-27	7		14			9.3		27.5		3
2023-03-27	7		27			40.9		31.9		3
2023-03-27	8		15			37.3		29.1		3
2023-03-27	8		24			11.0		25.5		3
2023-03-27	8		13			38.1		30.4		3
2023-03-27	8		63			10.1		27.3		3
2023-03-27	9		11			35.6		28.0		3
2023-03-27	9		59			7.0		24.7		3
2023-03-27	9		35			10.3		28.9		3
2023-03-27	9		31			42.4		33.3		3
2023-03-27	10		76			11.0		27.3		3
2023-03-27	10		NOPOT							3

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-03-27	10		28			38.3		30.1		3
2023-03-27	10		26			11.0		27.4		3
2023-03-27	11		12			36.5		27.5		3
2023-03-27	11		54			40.3		31.3		3
2023-03-27	11		6			8.8		30.7		3
2023-03-27	11		71			9.6		32.1		3
2023-03-27	12		44			10.2		30.0		3
2023-03-27	12		4			39.6		28.2		3
2023-03-27	12		60			8.0		29.5		3
2023-03-27	12		40			44.3		25.4		3
2023-03-27	13		17			8.2		29.3		3
2023-03-27	13		42			9.4		25.9		3
2023-03-27	13		3			40.2		23.2		3
2023-03-27	13		5			48.3		27.3		3
2023-03-27	14		20			38.9		25.4		3
2023-03-27	14		2			10.7		29.1		3
2023-03-27	14		19			10.1		26.3		3
2023-03-27	14		43			30.3		27.3		3
2023-03-27	15		67			5.1		28.8		3
2023-03-27	15		46			41.6		24.5		3

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-03-27	15		34			4.8		20.0		3
2023-03-27	15		23			36.5		27.4		3
2023-03-27	16		32			39.5		20.6		3
2023-03-27	16		69			7.9		29.3		3
2023-03-27	16		53			9.0		26.6		3
2023-03-27	16		21			39.4		22.7		3
2023-03-27	17		36			38.7		26.2		3
2023-03-27	17		80			12.5		29.6		3
2023-03-27	17		1			42.0		26.8		3
2023-03-27	17		18			11.7		26.9		3
2023-04-03	1		48			40.1		30.7		3
2023-04-03	1		64			8.8		34.1		3
2023-04-03	1		68			10.3		36.5		3
2023-04-03	1		45			44.6		23.3		3
2023-04-03	2		75			10.1		37.1		3
2023-04-03	2		51			41.0		23.2		3
2023-04-03	2		61			47.3		32.9		3
2023-04-03	2		57			11.7		33.2		3
2023-04-03	3		62			10.3		38.1		3
2023-04-03	3		33			41.0		31.1		3

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-04-03	3		52			9.7		39.8		3
2023-04-03	3		72			42.8		31.4		3
2023-04-03	4		70			8.4		37.7		3
2023-04-03	4		38			9.7		40.1		3
2023-04-03	4		74							3
2023-04-03	4		47			39.7		37.3		3
2023-04-03	5		73			40.6		37.0		3
2023-04-03	5		77			41.6		35.3		3
2023-04-03	5		79			12.0		44.9		3
2023-04-03	5		39			8.0		33.7		3
2023-04-03	6		65			42.7		27.2		3
2023-04-03	6		22			7.6		35.2		3
2023-04-03	6		58			9.4		42.4		3
2023-04-03	6		66			44.0		32.3		3
2023-04-03	7		25			39.9		27.0		3
2023-04-03	7		41			10.3		45.9		3
2023-04-03	7		14			9.3		40.4		3
2023-04-03	7		27			41.3		28.4		3
2023-04-03	8		15			37.5		40.5		3
2023-04-03	8		24			11.2		38.5		3

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-04-03	8	13				38.6		27.9		3
2023-04-03	8	63				10.3		39.4		3
2023-04-03	9	11				36.2		24.6		3
2023-04-03	9	59				7.0		34.8		3
2023-04-03	9	35				10.8		40.9		3
2023-04-03	9	31				42.9		43.2		3
2023-04-03	10	76				10.0		38.5		3
2023-04-03	10	NOPOT								3
2023-04-03	10	28				37.8		32.6		3
2023-04-03	10	26				11.1		41.9		3
2023-04-03	11	12				36.6		38.3		3
2023-04-03	11	54				39.5		33.1		3
2023-04-03	11	6				8.8		44.9		3
2023-04-03	11	71				9.6		44.2		3
2023-04-03	12	44				10.3		46.6		3
2023-04-03	12	4				40.2		30.4		3
2023-04-03	12	60				8.1		42.1		3
2023-04-03	12	40				45.9		28.8		3
2023-04-03	13	17				8.0		41.9		3
2023-04-03	13	42				8.5		45.2		3

date	block	row	pot_id	inoc_status	abiotic	height_cm	leaf_count	spad	sub	GH_zone
2023-04-03	13	3				40.3		31.1		3
2023-04-03	13	5				50.0		24.3		3
2023-04-03	14	20				40.2		31.7		3
2023-04-03	14	2				10.5		42.8		3
2023-04-03	14	19				10.4		40.2		3
2023-04-03	14	43				31.6		23.8		3
2023-04-03	15	67				4.9		36.5		3
2023-04-03	15	46				44.2		29.4		3
2023-04-03	15	34				5.1		32.6		3
2023-04-03	15	23				35.8		31.3		3
2023-04-03	16	32				41.1		26.2		3
2023-04-03	16	69				8.1		36.0		3
2023-04-03	16	53				9.8		36.2		3
2023-04-03	16	21				39.9		31.3		3
2023-04-03	17	36				38.2		35.9		3
2023-04-03	17	80				12.5		41.9		3
2023-04-03	17	1				42.8		28.3		3
2023-04-03	17	18				11.4		30.8		3

Check for any lost data

By number of rows

Number of rows should be equal.

```
is_equal<- nrow(df_combined_clean_327_403) == (nrow(df_230327_cleaned)+nrow(df_230403_cleaned))  
cat("The value of is_equal is", is_equal, ".")
```

The value of is_equal is TRUE .

Write new csv with the saved cleaned dataframe

```
write.csv(df_combined_clean_327_403,  
          "~/PowerAnalysis-Research/power_analysis/CleanedData/GH2211_nitrogenGrowth_327_403.csv")
```